

Power functions

Outline

Power functions

- Rational exponent

- Real exponent

- Polynomials

Direct and inverse proportionality functions

Power functions

Introduction: Power functions

- ▶ In this section we study the **power functions** $x \mapsto x^\alpha$, among the most important elementary functions
- ▶ We will see that the **domain**, the **symmetry**, the **sign** and the **monotonicity** depend on the exponent α (rational or real)
- ▶ Finally we introduce **polynomials**, linear combinations of powers with integer exponent



Definition 1: power functions

Given $\alpha \in \mathbb{R}, \alpha \neq 0$, the function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, $f : x \mapsto x^\alpha$ is called the **power function** with exponent α . The domain D depends on the value of α :

- ▶ with rational exponent $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have four cases:

$$D = \begin{cases} \mathbb{R} & \text{if } n \text{ odd and } \frac{m}{n} > 0 \\ \mathbb{R} \setminus \{0\} & \text{if } n \text{ odd and } \frac{m}{n} < 0 \\ [0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} > 0 \\ (0, +\infty) & \text{if } n \text{ even and } \frac{m}{n} < 0 \end{cases}$$

- ▶ with real, non-rational exponent $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, we have two cases: $D = [0, +\infty)$ if $\alpha > 0$ and $D = (0, +\infty)$ if $\alpha < 0$.

Rational exponent

With $\alpha = \frac{m}{n} \in \mathbb{Q}$, $m \in \mathbb{Z}$ and $n \in \mathbb{N}_+$ coprime, we have a power function with rational exponent:

$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto x^{\frac{m}{n}} = \sqrt[n]{x^m}$$

We have the following three cases, depending on whether m and n are even or odd.

Case 1: n odd and m even (1/2)

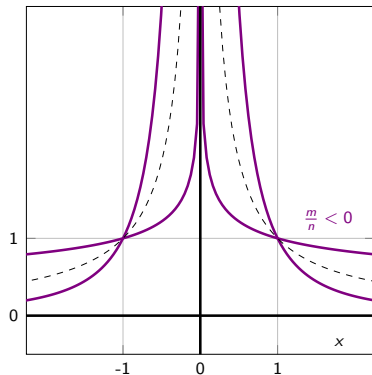
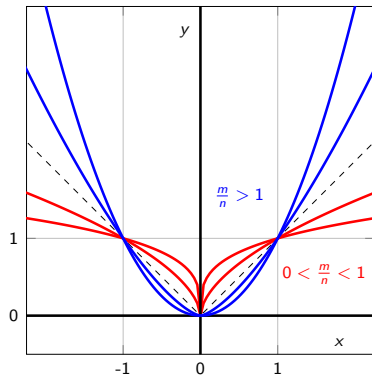
With n odd and m even, it is an **even function**, i.e., $(-x)^{\frac{m}{n}} = x^{\frac{m}{n}}$.

For the sign and the monotonicity we have:

$$x^{\frac{m}{n}} > 0, \quad \forall x \in \mathbb{R} \setminus \{0\}; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \quad \text{and} \quad \frac{m}{n} > 0$$

- ▶ if $\frac{m}{n} > 0$ then f is decreasing in $(-\infty, 0]$ and increasing in $[0, +\infty)$
- ▶ if $\frac{m}{n} < 0$ then f is increasing in $(-\infty, 0)$ and decreasing in $(0, +\infty)$

Case 1: n odd and m even (2/2)



Case 2: n odd and m odd (1/2)

With n odd and m odd, it is an **odd function**, i.e., $(-x)^{\frac{m}{n}} = -x^{\frac{m}{n}}$.

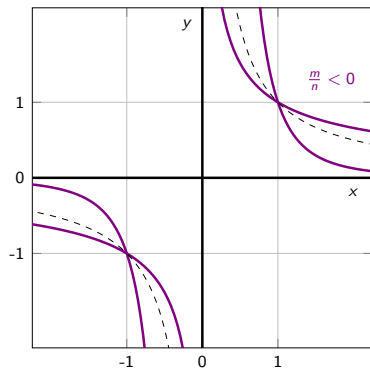
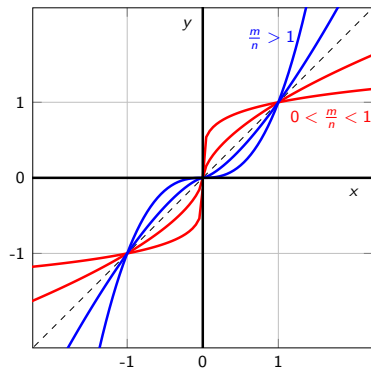
For the sign and the monotonicity we have:

$$x^{\frac{m}{n}} < 0, \quad \forall x < 0, \quad x^{\frac{m}{n}} > 0, \quad \forall x > 0;$$

$$x^{\frac{m}{n}} = 0 \iff x = 0 \quad \text{and} \quad \frac{m}{n} > 0$$

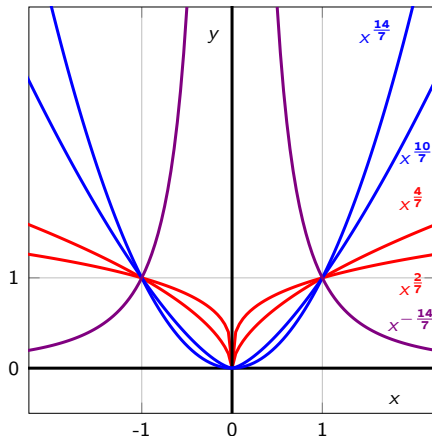
- ▶ if $\frac{m}{n} > 0$ then f is increasing in $\mathbb{R}$
- ▶ if $\frac{m}{n} < 0$ then f is decreasing in $(-\infty, 0)$ and in $(0, +\infty)$

Case 2: n odd and m odd (2/2)



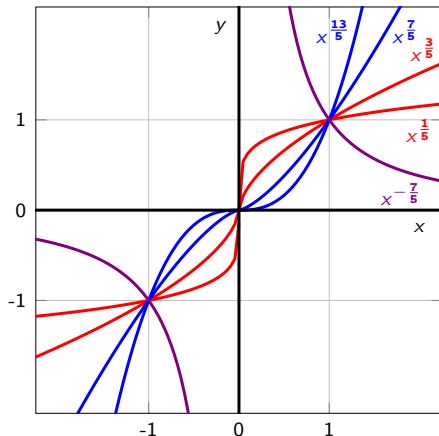
Example (1/2): graphs of power functions with rational exponent $\frac{m}{n}$ and n odd

With m even:



Example (2/2): graphs of power functions with rational exponent $\frac{m}{n}$ and n odd

With m odd:



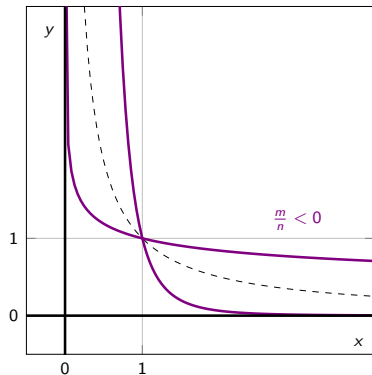
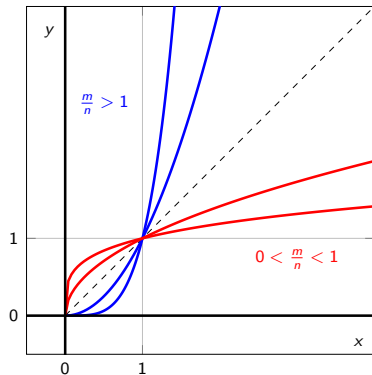
Case 3: n even and m odd (1/2)

With n even and m odd (m cannot be even since they are coprime), for the sign and the monotonicity we have:

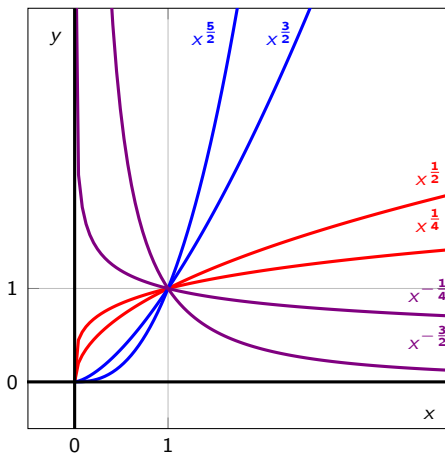
$$x^{\frac{m}{n}} > 0, \quad \forall x > 0; \quad x^{\frac{m}{n}} = 0 \iff x = 0 \quad \text{and} \quad \frac{m}{n} > 0$$

- ▶ if $\frac{m}{n} > 0$ then f is increasing in $[0, +\infty)$
- ▶ if $\frac{m}{n} < 0$ then f is decreasing in $(0, +\infty)$

Case 3: n even and m odd (2/2)



Example: graphs of power functions with rational exponent $\frac{m}{n}$ and n even



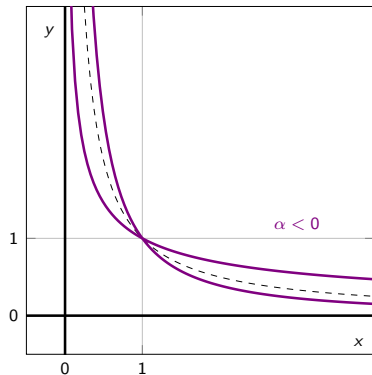
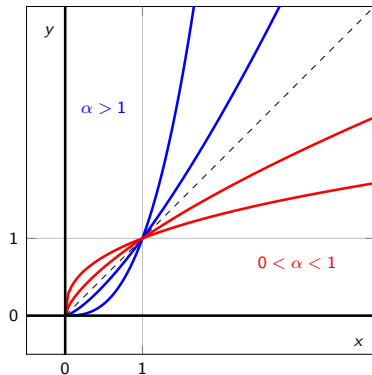
Real exponent (1/2)

- ▶ With $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, for the sign and the monotonicity we have:

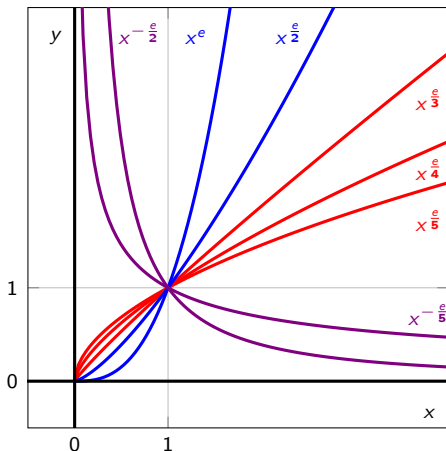
$$x^\alpha > 0, \quad \forall x > 0; \quad x^\alpha = 0 \iff x = 0 \quad \text{and} \quad \alpha > 0$$

- ▶ if $\alpha > 0$ then f is increasing in $[0, +\infty)$
- ▶ if $\alpha < 0$ then f is decreasing in $(0, +\infty)$

Real exponent (2/2)



Example: graphs of power functions with real exponent



Definition 2: polynomial

Given $n + 1$ values $a_i \in \mathbb{R}$ with $i \in \{0, 1, \dots, n\}$ and $a_n \neq 0$, the function:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \underbrace{\sum_{i=0}^n a_i x^i}_{=P_n(x)}$$

is called a **polynomial** of degree n .

Monomials

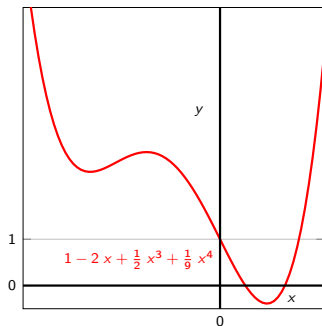
- ▶ For each **monomial** $i \in \{0, 1, \dots, n\}$:
 1. the value a_i is the **coefficient of the monomial**
 2. the power function x^i with integer exponent i is the **literal part** of the monomial

The value a_0 is the **constant term** of the polynomial since $x^0 = 1$.

Example: polynomial

For example, with $n = 4$, $a_0 = 1$, $a_1 = -2$, $a_2 = 0$, $a_3 = \frac{1}{2}$ and $a_4 = \frac{1}{9}$ we have the following polynomial of degree four:

$$P_4(x) = \sum_{i=0}^4 a_i x^i = 1 - 2x + \frac{1}{2}x^3 + \frac{1}{9}x^4$$



Summary

Power functions $x \mapsto x^\alpha$:

- ▶ The domain depends on α : with $\alpha = \frac{m}{n}$, on the parity of n and on the sign of α ; with irrational α , $D = [0, +\infty)$ or $(0, +\infty)$
- ▶ n odd, m even: **even** function; n odd, m odd: **odd** function
- ▶ With $\alpha > 0$ the power is increasing on $[0, +\infty)$, with $\alpha < 0$ decreasing on $(0, +\infty)$

Polynomials:

- ▶ $P_n(x) = \sum_{i=0}^n a_i x^i$ with $a_n \neq 0$: sum of monomials with coefficients a_i ; a_0 is the constant term

Direct and inverse proportionality functions



Introduction: Direct and inverse proportionality functions

- ▶ In this section we study what happens when a power function is multiplied by a **constant** λ
- ▶ The cases $\alpha = 1$ and $\alpha = -1$ give the **direct proportionality** functions (lines through the origin) and the **inverse proportionality** functions (rectangular hyperbolas)
- ▶ We will see how the value and the sign of λ change the graph

Direct proportionality

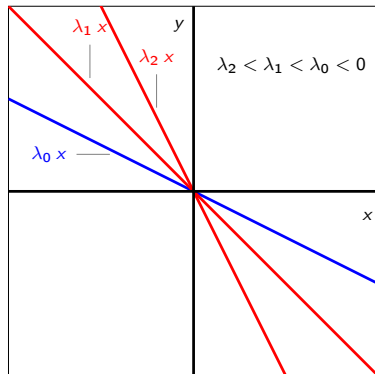
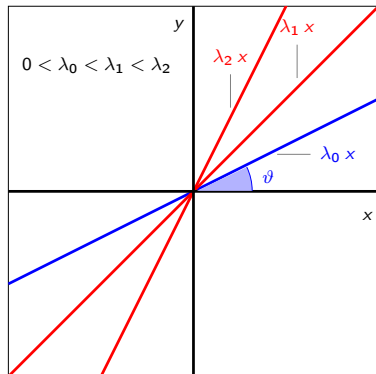
With $\alpha = 1$ and given $\lambda \in \mathbb{R}$, we have the family of **direct proportionality (or linear) functions**:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \lambda x$$

where λ is the constant of direct proportionality.

Moreover, $\lambda = \tan \vartheta$ and ϑ is the angle between the line $y = \lambda x$ and the x -axis.

Direct proportionality: graphs



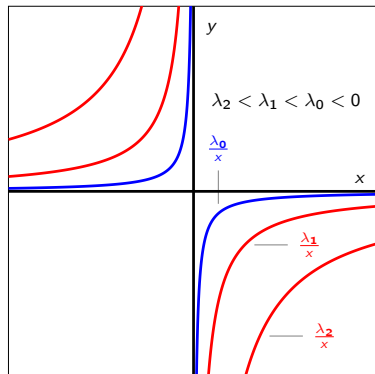
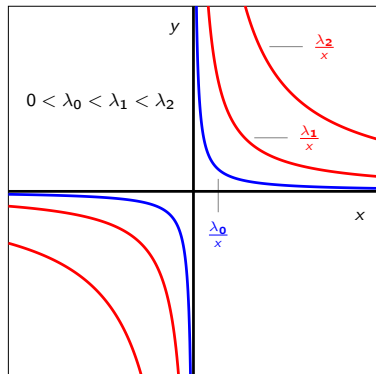
Inverse proportionality

With $\alpha = -1$ and given $\lambda \in \mathbb{R}$, we have the family of **inverse proportionality functions (or rectangular hyperbolas)**:

$$f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}, \quad f : x \mapsto \frac{\lambda}{x}$$

where λ is the constant of inverse proportionality.

Inverse proportionality: graphs



Powers multiplied by a constant

Given $\alpha \in \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have the family of **power functions multiplied by a constant**:

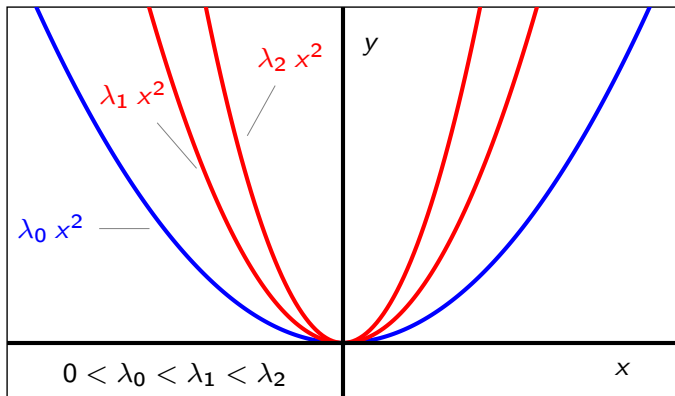
$$f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}, \quad f : x \mapsto \lambda x^\alpha$$

where λ is the multiplicative constant.

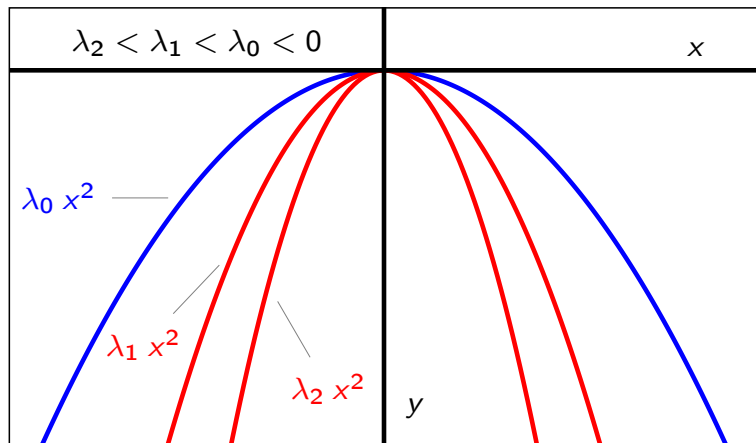
- We now see some examples with $\alpha = 2$, $\alpha = 3$ and $\alpha = \frac{1}{2}$.

Example (1/5): $\alpha = 2$

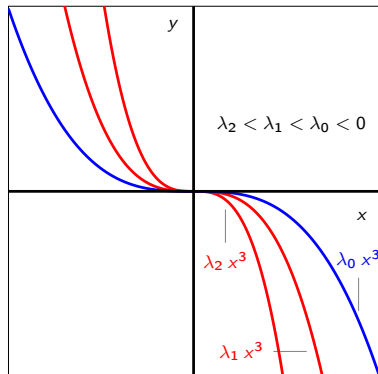
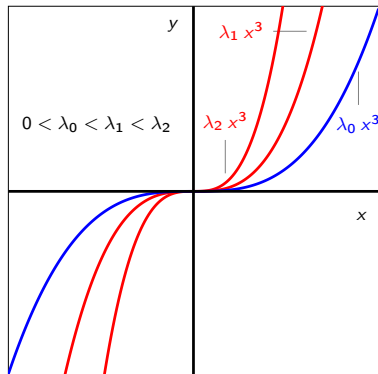
- For example, with α equal to 2 or equal to 3 and $\lambda \in \mathbb{R}$ we have:



Example (2/5): $\alpha = 2$

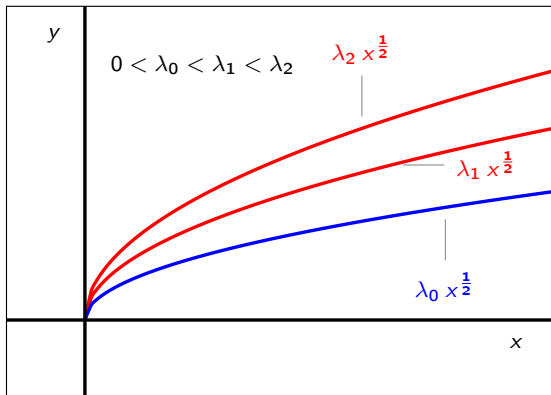


Example (3/5): $\alpha = 3$

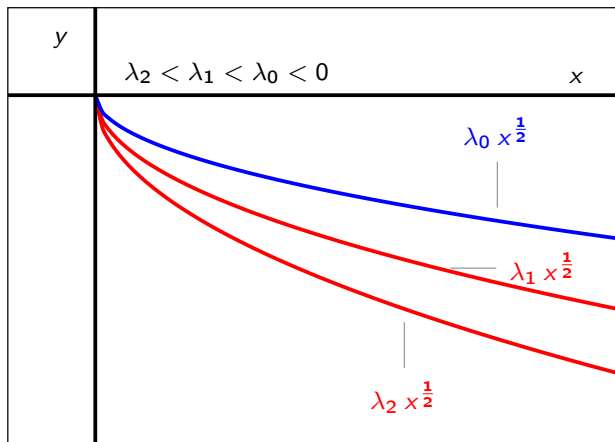


Example (4/5): $\alpha = \frac{1}{2}$

- For example, with $\alpha = \frac{1}{2}$ and $\lambda \in \mathbb{R}$ we have:



Example (5/5): $\alpha = \frac{1}{2}$



Direct and inverse proportionality:

- ▶ $\alpha = 1$: $x \mapsto \lambda x$, lines through the origin with $\lambda = \tan \vartheta$
- ▶ $\alpha = -1$: $x \mapsto \frac{\lambda}{x}$, rectangular hyperbolas
- ▶ In general $x \mapsto \lambda x^\alpha$: the constant λ stretches/compresses the graph and, if negative, reflects it with respect to the x -axis

Final summary

- ▶ **Power functions** x^α have domain, symmetry and monotonicity depending on the exponent α
- ▶ Rational exponent $\frac{m}{n}$: three cases depending on the parity of m and n ; irrational exponent: domain contained in $[0, +\infty)$
- ▶ **Polynomials** are sums of monomials $a_i x^i$
- ▶ The functions λx (**direct proportionality**), $\frac{\lambda}{x}$ (**inverse proportionality**) and λx^α show the effect of a multiplicative constant